Considerations for Underwater ROV Design

Designing Remotely Operated Vehicles and subsea tools demands careful consideration of several factors, including corrosion protection, material selection, and surface finish.

The propeller is Aluminum Alloy

Straws that use a filter of sorts on the outside to block out unwanted debris, and allow for water to be sucked in.

PROBLEM STATEMENT

Mary Kerr has requested that an ROV be assembled to collect water microplastic samples of about 5 mm for a marine bio class project. Mr. Infante will be providing materials and instructing us through our journey of creating this ROV., his primary concern is creating a machine successful in capturing various samples of microplastics. Our findings can lead to proactive initiatives to solve the microplastic problem by working alongside San Francisco Mayor Daniel Lurie. The Marine Bio class needs the microplastic samples, through which they will conduct their testing on plastics and how they affect marine life. Marine animals will also be affected by this project; the sampling and research on the microplastics could

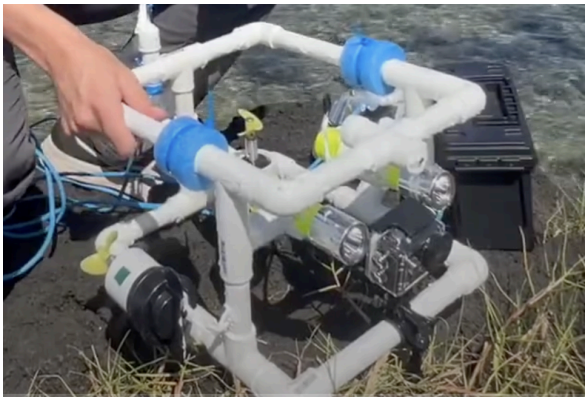
lead to a better understanding of how to prevent further spreading of them. The testing for this project is planned for the end of March to mid-April. This mission aims to understand and solve the microplastic problem that plagues the ocean due to humans. Microplastics are synthetic and high molecular compounds typically sized around 5mm. There is a multitude of microplastics; the most common are polyethylene, polypropylene, and polystyrene. These plastics are a result of the broken-up structures of other larger plastic products, which have been broken up so much that they are very small and extremely difficult to get rid of. This is a recurring problem that the world has been dealing with since its discovery in the 1960s.

Research to Start the Build (01/27) -

- Bigger is not always better; keep the ROV symmetrical, and understand insertion length.
- Rectangular boxed ROVs seem to be the most common design, including two layers of PVC, and cuts of pool noodles positioned at the center of the longest side.
- Motors work best on either the side or the back of the ROV(motors pointed up or down, allowing for omnidirectional movement.
- The camera is positioned in the center of the front.
- Ask yourself what each piece of PVC pipe is doing for you. (NO WASTE)

- Set screws instead of glue for the PVC pipes.
- The ROV frame should expect to be flooded with water considering we are incapable of forming an air-tight seal. * Tees extensions at the corners will allow water to flow through.
- Heavier items should be kept around the center point of the ROV.

References (Images) -



[YT Video for Reference]